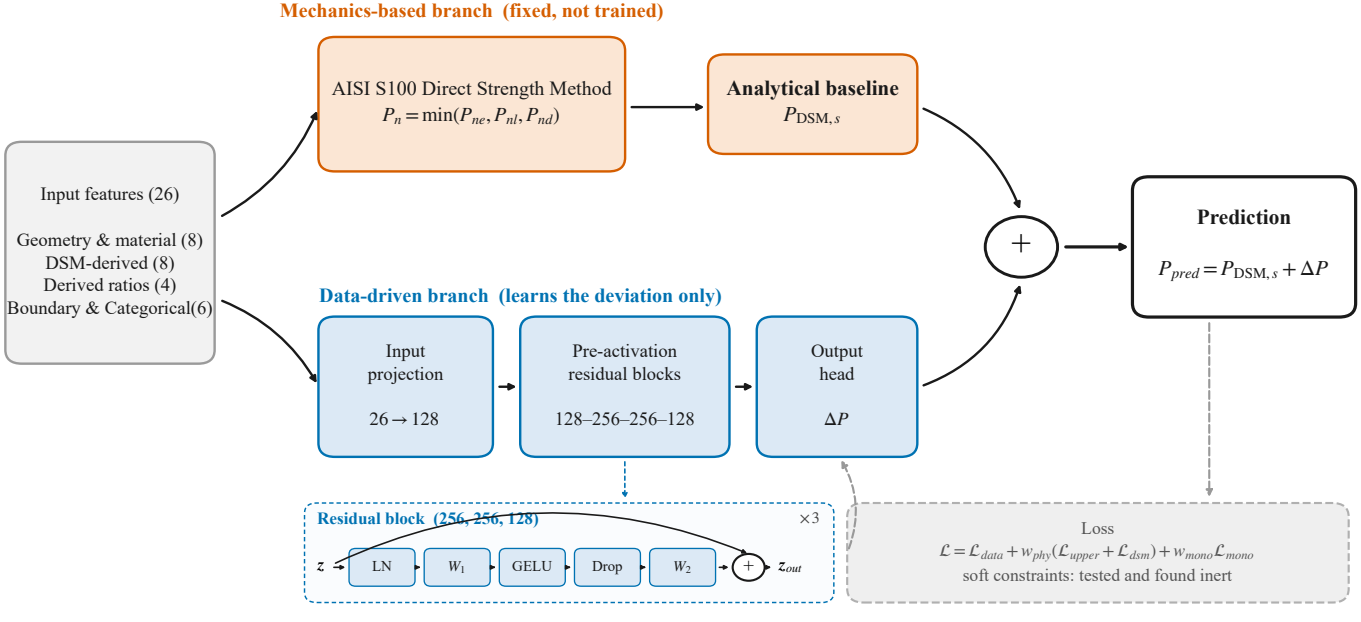

Mechanics-based branch (fixed, not trained)
AISI S100 Direct Strength Method
$P_n = \min(P_{ne}, P_{nl}, P_{nd})$
Analytical baseline
$P_{\mathrm{DSM},s}$
Input features (26)
Geometry & material (8)
DSM-derived (8)
Derived ratios (4)
Boundary & Categorical(6)
Data-driven branch (learns the deviation only)
Input projection
26 → 128
Pre-activation residual blocks
128–256–256–128
Output head
ΔP
+
Prediction
$P_{pred} = P_{\mathrm{DSM},s} + \Delta P$
Residual block (256, 256, 128)
×3
z
LN
W_1
GELU
Drop
W_2
+
z_{out}
Loss
$\mathcal{L} = \mathcal{L}_{data} + w_{phy}(\mathcal{L}_{upper} + \mathcal{L}_{dsm}) + w_{mono}\mathcal{L}_{mono}$
soft constraints: tested and found inert